

KOSH-AGACH, Russian Federation

WMO: 362590

Lat: 50.0088N

Lon: 88.6721E

Elev: 1760

StdP: 81.89

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.8	-38.5	-43.8	0.1	-40.8	-41.3	0.1	-38.3	7.6	-10.9	5.7	-17.2	0.3	90	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	25.1	12.1	23.5	11.5	22.0	11.1	13.7	21.7	12.8	19.9	12.1	19.3	2.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.2	10.3	14.9	10.2	9.6	14.8	9.2	9.0	14.3	44.3	21.7	41.7	20.1	39.6	19.3	18.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	7.9	6.4	DB	-41.4	28.3	3.1	1.5	-43.7	29.4	-45.5	30.3	-47.3	31.2	-49.6	32.3	
			WB	-41.5	15.2	3.1	1.7	-43.7	16.5	-45.6	17.5	-47.3	18.5	-49.6	19.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.5	-27.6	-23.4	-10.2	1.8	6.6	14.0	15.4	13.3	6.6	-1.7	-14.2	-23.7
	DBStd	15.98	6.37	6.49	7.75	4.71	4.00	3.39	2.67	2.90	4.21	5.35	7.52	5.60
	HDD10.0	5359	1167	937	628	246	117	7	1	7	114	364	725	1046
	HDD18.3	7978	1425	1170	886	496	363	133	96	158	351	622	975	1304
	CDD10.0	431	0	0	0	0	13	128	168	108	13	0	0	0
	CDD18.3	8	0	0	0	0	0	3	4	0	0	0	0	0
	CDH23.3	153	0	0	0	0	1	58	73	19	3	0	0	0
	CDH26.7	12	0	0	0	0	0	6	6	0	0	0	0	0
	WSAvg	1.8	0.8	1.0	1.6	2.5	3.1	2.3	2.0	1.9	1.9	1.8	1.4	1.0
Precipitation	PrecAvg	115	3	2	2	3	8	20	30	22	10	3	5	6
	PrecMax	187	18	8	7	18	37	57	69	48	31	12	22	22
	PrecMin	49	0	0	0	0	0	2	2	2	0	0	0	0
	PrecStd	31	4	2	2	4	7	12	16	12	7	3	6	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.7	-4.0	8.8	16.8	22.1	27.3	27.4	25.2	23.2	14.7	5.1	-5.8
		MCWB	-10.3	-7.2	1.5	6.1	8.7	13.3	12.9	11.9	10.0	5.6	1.1	-7.4
	2%	DB	-11.7	-7.4	6.1	14.3	18.9	25.0	25.4	23.7	19.7	11.7	1.9	-9.1
		MCWB	-13.0	-9.5	0.1	5.0	7.4	11.7	12.8	11.6	8.9	4.0	-0.8	-10.3
	5%	DB	-14.8	-10.3	3.8	11.9	16.8	23.1	23.4	22.2	17.1	9.0	-1.1	-12.3
		MCWB	-15.7	-11.8	-1.2	3.8	6.7	10.9	12.2	11.0	7.8	2.9	-3.5	-13.2
	10%	DB	-18.4	-13.4	1.2	9.9	14.9	21.4	21.9	20.6	14.8	6.6	-3.8	-15.4
		MCWB	-19.0	-14.5	-2.4	2.6	6.2	10.5	11.6	10.7	6.6	1.5	-5.7	-16.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.2	-6.8	2.2	6.8	9.6	14.7	15.8	14.0	10.7	6.1	1.9	-7.2
		MCDB	-8.9	-4.3	7.6	14.9	18.7	24.8	23.3	21.8	19.4	13.1	3.7	-6.2
	2%	WB	-12.9	-9.4	0.6	5.7	8.5	13.1	14.3	12.8	9.4	4.5	-0.7	-10.3
		MCDB	-11.6	-7.4	5.5	12.8	17.1	21.1	21.9	19.9	17.8	10.6	2.0	-9.2
	5%	WB	-15.8	-11.8	-0.9	4.5	7.5	12.3	13.2	12.0	8.4	3.1	-3.5	-13.1
		MCDB	-14.8	-10.4	3.2	10.7	15.8	19.8	20.2	19.4	16.1	8.6	-1.3	-12.2
	10%	WB	-19.0	-14.5	-2.3	3.2	6.4	11.5	12.4	11.3	7.2	1.6	-5.7	-16.1
		MCDB	-18.4	-13.4	1.4	8.9	13.7	18.8	19.3	18.6	13.9	6.3	-3.8	-15.4
Mean Daily Temperature Range	5% DB	MDBR	10.5	12.6	13.0	12.0	11.9	11.8	11.2	12.3	13.2	11.8	10.2	9.6
		MCDBR	12.4	13.6	13.4	15.3	15.3	14.7	14.0	15.3	16.8	15.9	11.5	11.2
		MCWBR	11.3	11.8	8.4	7.6	7.0	4.8	4.7	5.5	7.6	9.1	8.7	10.2
	5% WB	MCDBR	12.5	13.5	12.5	13.8	13.3	12.4	12.0	12.9	15.2	15.0	11.1	11.4
		MCWBR	11.5	11.7	8.2	7.1	6.5	4.4	4.7	5.0	7.0	8.8	8.5	10.5
Clear-Sky Solar Irradiance	taub		0.228	0.253	0.301	0.348	0.341	0.341	0.340	0.324	0.289	0.277	0.251	0.230
	taud		2.312	2.265	2.197	2.213	2.299	2.391	2.412	2.429	2.480	2.434	2.354	2.295
	Ebn at Noon		846	901	903	893	915	916	910	911	916	864	803	783
	Edh at Noon		65	89	116	129	124	114	110	103	87	75	62	57
All-Sky Solar Radiation	RadAvg		1.48	2.51	3.99	5.37	6.29	6.26	5.99	5.34	4.16	2.59	1.50	1.14
	RadStd		0.07	0.09	0.18	0.23	0.33	0.34	0.28	0.26	0.29	0.22	0.09	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	-1.43	-1.63	N/A	N/A	N/A	N/A

Nomenclature: See separate page